



NON POLAR MOLECULES

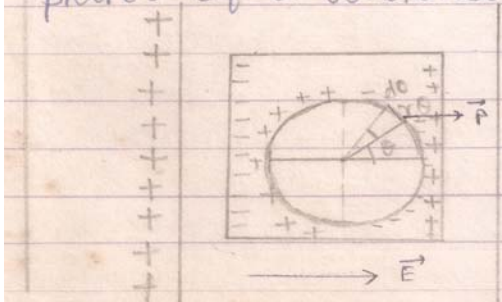
Clausius-Mosotti equation:

Lorentz local field: In a gas the molecules are at relatively large distance so that the mutual interaction between the molecules may be neglected when the gaseous dielectric is placed in an electrostatic field and the only force acting on the molecule is that due to the externally applied field. However in case of solid or liquid dielectric the atoms or molecules not only experience force due to external field but also the force produced by the induced dipoles and is known as dipole-dipole interaction.

Thus the actual field responsible for polarizing a dielectric is not the externally applied field E but a lesser field E known as local field or internal field or effective field. Hence the induced dipole moment $m = \alpha E_0 F$ — (1)

The local field F is also known as Lorentz local field.

To evaluate Lorentz local field let us consider non-polar molecules of a homogeneous isotropic medium lying between the parallel plates of a condenser:



let us consider a small spherical region inside the dielectric (the region is microscopic compared to the size of the dielectric but large compared to



the size of the molecule).

We want to calculate the Lorentz local field F acting on a molecule by considering the force on the unit positive charge at the centre of the spherical region, which consists of following parts.

- (1) F_1 the force due to the charge density σ_f on the parallel conducting plates on either side of the dielectric which is given by

$$F_1 = \frac{\sigma_f}{\epsilon_0}$$

- (2) F_2 the force due to induced charge density ' σ_p ' on the surfaces of the dielectric adjacent to the plates which is given by

$$F_2 = - \frac{\sigma_p}{\epsilon_0}$$

The negative sign is due to the field produced by polarization charges being opposite to that produced by charges on the plates.

- (3) F_3 , the force due to the induced charges on the surface of the small spherical surface (fig)

- (4) F_4 ; the force due to polarization of the material within the sphere which is zero for gases & non-associated liquids in which the molecules are oriented at random in the



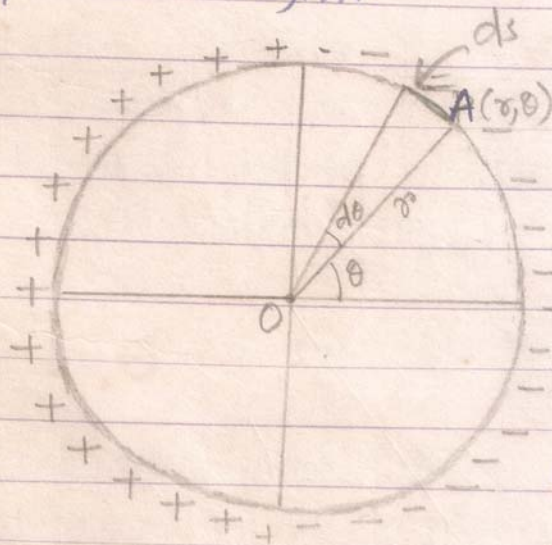
absence of an externally applied field. It is also zero in the case of cubic crystals. We shall ignore it here.

Therefore F acting on a unit positive charge (which is equal to electric intensity E at the centre of the spherical region) is given by:

$$F = F_1 + F_2 + F_3$$

$$F = \frac{\sigma_t}{\epsilon_0} - \frac{\sigma_p}{\epsilon_0} + F_3$$

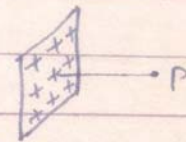
Evaluation of F_3 : We now calculate the intensity at a point O , due to the induced charge on the surface of the spherical region.



Let r = radius of the spherical region.

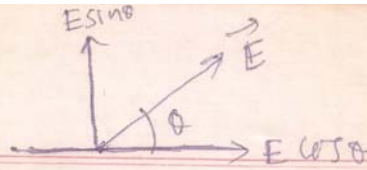
Let us consider a point A on the surface of the spherical region having co-ordinate (r, θ)

At ' A ' let us consider an element of area ds on the surface of the spherical region.



from the app. of Gauss's theorem: Int.

at a point near an infinitely charged conducting plate = $\frac{\sigma}{\epsilon} = E$



vector $\vec{P}$ is along the direction of the applied field $\vec{E}$.

The component of the polarization vector along normal to the surface of the spherical region
 $= P \cos \theta =$ the surface density of the induced charge in the dielectric.

$\therefore$ The charge on the element $= (P \cos \theta) ds$.

Electric intensity at O, due to this charge

$$E = \frac{P \cos \theta ds}{4\pi\epsilon_0 r^2} \text{ along } \vec{OA} \text{ [because of } (-) \text{ve charge]}$$

The component of this intensity along the direction of the externally applied field $= E \cos \theta$

$$= \frac{P \cos \theta ds}{4\pi\epsilon_0 r^2} \cdot \cos \theta$$

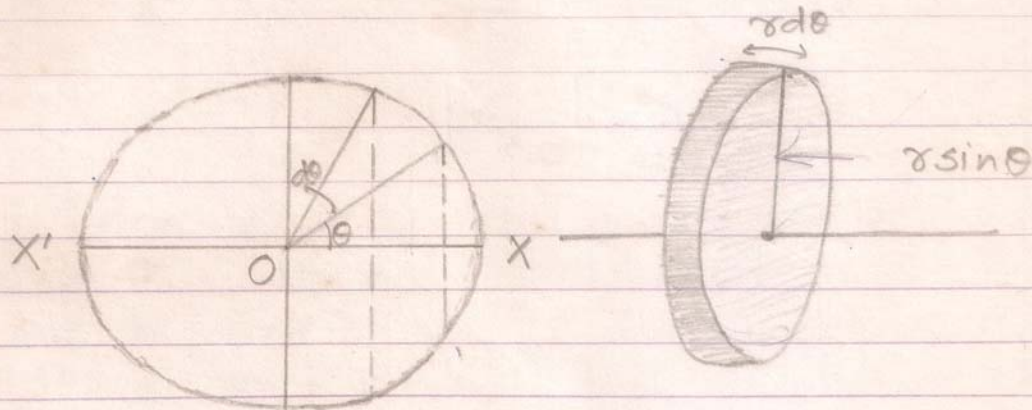
The component along perpendicular to the direction of the applied field $= E \sin \theta$

$$= \frac{P \cos \theta ds}{4\pi\epsilon_0 r^2} \cdot \sin \theta$$

Since this component perpendicular to the direction of the applied field due to the charge on the surface elements on the 1st and 4th quadrants are opposite in direction; when added over all the elements in the right hemisphere will mutually cancel out and the component in the



direction of the field will add up.



Let us consider an elementary disc, cut by two planes perpendicular to the axis XX' .
The area of the surface at the edge of this elementary disc = Circumference $\times$ thickness
 $ds = 2\pi (r \sin \theta) (r d\theta)$

The Total intensity at O , in the direction of the externally applied field XX' due to the charge present on the edge of this elementary disc = $\frac{P \cos^2 \theta}{2\epsilon_0} \cdot 2\pi r^2 \sin \theta d\theta$
$$= \frac{P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0}$$

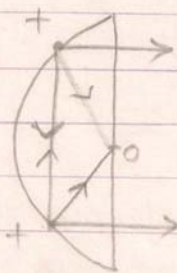
Hence the total electric intensity at O , in the direction of the externally applied field due to the charge on the surface of the right hemisphere

$$I = \int_0^{\pi/2} \frac{P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0}$$



putting $\cos\theta = x \therefore -\sin\theta d\theta = dx$
 when $\theta = 0$; $x = 1$; $\theta = \pi/2$, $x = 0$

$$\therefore I = -\frac{P}{2\epsilon_0} \left[\frac{x^3}{3} \right]_1^0 = \frac{P}{6\epsilon_0}$$



Since the surface density of charge on the +ve and -ve hemisphere are equal and opposite;

We find that the total electric intensity at O in the direction of the field due to the entire charge on the left half of the hemisphere will also be $\frac{P}{6\epsilon_0}$

$\therefore$ The total electric intensity at O ;

$$F_3 = \frac{2P}{6\epsilon_0} = \frac{P}{3\epsilon_0}$$

The total force F on the unit positive charge at O due to all causes (i.e. the actual intensity of electric field) is then given by

$$F = \frac{\sigma_t}{\epsilon_0} - \frac{\sigma_p}{\epsilon_0} + \frac{P}{3\epsilon_0}$$

But the intensity of electric field 'E' inside a dielectric is given by

$$E = \frac{\sigma_t - \sigma_p}{\epsilon_0}$$



Clausius - Mosotti eqⁿ holds good only for liquids & gases and for ^{nm} polar molecules.

$$\therefore F = E + \frac{P}{3\epsilon_0}$$

Clausius - Mosotti Equation :

In clausius-mosotti theory we consider the actual non-polar molecules of a homogenous isotropic dielectric (and not the dielectric as a continuous medium) laying between the plates of a parallel-plate condenser which are large compared to the distance between them. The Clausius - Mosotti equation gives the dielectric constant, in terms of molecular properties.

We have seen above that when a non-polar molecule is placed in a electric field, the actual intensity of electric field (i.e Lorentz local field) acting on any molecule is given by:

$$F = E + \frac{P}{3\epsilon_0}$$

The dipole moment induced in a molecule
 $m = \alpha \epsilon_0 F = \alpha \epsilon_0 \left(E + \frac{P}{3\epsilon_0} \right)$

The dipole moment per unit volume = $nm = P$

$$P = n \alpha \epsilon_0 \left(E + \frac{P}{3\epsilon_0} \right)$$

Where n is the number of molecules per unit volume.

Since $D = \epsilon_0 E + P$ $\therefore P = D - \epsilon_0 E$ putting this in the above equation we get :



$$\epsilon_r = n^2$$

$$n = \sqrt{\epsilon_r}$$

$$\begin{aligned} D - \epsilon_0 E &= n \alpha \epsilon_0 \left(E + \frac{D - \epsilon_0 E}{3 \epsilon_0} \right) \\ &= n \alpha \epsilon_0 \left(\frac{2E}{3} + \frac{D}{3 \epsilon_0} \right) = n \alpha \left(\frac{D + 2 \epsilon_0 E}{3} \right) \end{aligned}$$

$$\text{or, } \frac{D - \epsilon_0 E}{D + 2 \epsilon_0 E} = \frac{n \alpha}{3} \quad \text{But } D = \epsilon_r E$$

$$\text{or, } \frac{\epsilon_r - \epsilon_0}{\epsilon_r + 2 \epsilon_0} = \frac{n \alpha}{3} \quad \text{--- (1) putting } \epsilon_r = \epsilon_0 \epsilon \text{ we get}$$

$$\text{or, } \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{n \alpha}{3} \quad \text{--- (2)}$$

The eqⁿ (1) or (2) is the simplest form of Clausius-Mosotti equation and gives the permittivity or dielectric constant in terms of the polarisability of the dielectric. The function $\frac{\epsilon_r - \epsilon_0}{\epsilon_r + 2 \epsilon_0}$ (or $\frac{\epsilon_r - 1}{\epsilon_r + 2}$) is referred

to as volume polarisation and depends only on the number of molecules per unit volume and their polarisabilities. It is dimensionless and is denoted by π .

$$\pi = \frac{\epsilon_r - \epsilon_0}{\epsilon_r + 2 \epsilon_0} = \frac{n \alpha}{3} \quad \text{--- (3)}$$

putting $\epsilon_r = 1 + \chi$ the Clausius-Mosotti eqⁿ can be written in terms of the susceptibility of the dielectric.

$$\frac{\chi}{\chi + 3} = \frac{n \alpha}{3} \quad \text{--- (4)}$$



$$p_m = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} V_M = \frac{M Z \alpha}{3} \quad V_M = \text{molecular volume}$$

$$p_m = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \cdot V_M = \frac{3}{N} \frac{N \alpha}{3}$$

Multiplying both sides of eqⁿ (2) by sp. volume v , which is reciprocal of density d , we get

$$p = \pi v = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \cdot \frac{1}{d} = \frac{1 \cdot n \alpha v}{3}$$

$$p = \frac{Z \alpha}{3} \quad \text{--- (5) where } [Z = n v] \text{ is the number of molecules per unit mass.}$$

The quantity 'p' is referred to as Specific polarization. Further multiplying both sides of eqⁿ (5) by molecular weight M of the substance we get

$$p_m = p M = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \frac{M}{d} = \frac{1}{3} \frac{M Z \alpha}{d}$$

$$p_m = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} V_M = \frac{N \alpha}{3} \quad \text{--- (6) } V_M = \frac{M}{d}$$

where V_M is the molecular weight volume and N is the Avogadro's number. The

quantity p_m is referred to as molar polarization.

The quantities p , p_m and α have dimensions of volume.

If we put $\epsilon = \epsilon_0 n^2$ in eqⁿ (6) where n is the refractive index of the medium then we get:

$$\frac{n^2 - 1}{n^2 + 2} V_M = \frac{N \alpha}{3} \quad \text{This is known as Lorentz-Lorentz eqⁿ of light which was derived by them from electromagnetic theory of consideration.}$$

Discussion:-

The Clausius-Mosotti equation² shows that ϵ_r must increase rapidly and tend to infinity as the number of molecules per unit volume approaches the critical value η_c where

$$\eta_c \alpha = 1.$$

Physically,³ an infinite ϵ_r corresponds to the dielectric vol. filled with infinitely polarisable i.e. conducting molecules. In the case of gases and liquids this condition is never approached as the polarisabilities are relatively small. The Clausius-Mosotti equation is therefore valid in the case of gases and liquids. In case of crystalline solids, however the interactions of the dipoles with each other give rise to more complicated polarisations than those in the simple model and the Clausius-Mosotti equation does not valid.